Test Plan

Outside-In Due Diligence Tool

Workforce analysis + SEC EDGAR financial data for M&A engagements

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SCOPE — THREE CORE FLOWS

- **Flow 1 — Roster Analysis:** Excel upload or clipboard paste, headcount and labor spend KPIs, breakdowns by function, seniority and country.
- **Flow 2 — Financial Data:** automatic retrieval via SEC EDGAR or manual entry of Revenue, EBITDA and Expenses.
- **Flow 3 — Integrated Analysis:** cross-dataset KPIs, value creation opportunities, PowerPoint export.

1. Introduction

1.1 Purpose

This document describes the testing strategy for the **Outside-In Due Diligence Tool**, an application that allows practitioners to upload workforce data (LinkedIn export) and retrieve financial data from SEC EDGAR 10-K filings, combining both into an integrated analysis for M&A engagements.

1.2 Out of Scope

- Infrastructure and DevOps testing (Docker, Nginx, SQL Server)
- Security and penetration testing
- Load / stress testing
- Browser versions older than 2023
- Mobile devices

2. Tech Stack Relevant to QA

Layer	Technology
Frontend	React 19 / TypeScript / Vite (production) · Alpine.js + Chart.js (prototype)
Backend	Python FastAPI (prototype) · PHP Laravel 13 (production)
Database	SQL Server 2022
Automation	**Playwright**
External API	SEC EDGAR (data.sec.gov)

3. Testing Types

Type	Description
Functional	Validation of business flows, KPIs and navigation rules. Reference: `test-cases-happy-path.md` (TC-001 to TC-020).
E2E Automation	Playwright — critical happy path, calculations, business rules. See section 5.

API	Direct endpoint validation: `POST /api/roster/analyze`, `GET /api/financials/{ticker}`.
Exploratory	Free-form sessions to discover unspecified behaviors, especially in file parsing and AI narrative generation.
Cross-browser	Verification across the browsers expected to be used by AlixPartners practitioners.

4. Test Environments

Environment	Description	URL
Local (prototype)	FastAPI + static HTML	`http://localhost:8000`
Local (production)	Laravel + React in Docker	`http://localhost`
Staging	TBD — to be defined with the team	—

5. Playwright Automation Strategy

5.1 Selection Criteria

A test case is automated if it meets at least two of the following criteria:

- It is part of the critical happy path
- It involves a numeric calculation that can be precisely asserted
- It is a business rule that can break silently after a refactor
- It runs in every regression cycle

5.2 Cases that WILL be automated

TC	Case	Reason
TC-001	Excel file upload	Critical happy path; Playwright supports `setInputFiles()` natively
TC-003	Headcount and cost KPIs	Numeric calculations that can be precisely asserted
TC-004	Senior ratio and cost concentration KPIs	Business rules that can break silently
TC-005	Headcount by function	Total consistency — straightforward DOM assertions
TC-006	Headcount by seniority	Hierarchy order + totals verifiable
TC-007	Labor spend by function	Numeric totals verifiable
TC-008	Geographic breakdown table	Table structure verifiable with `getByRole`
TC-011	Manual financial data entry	Form with validation — strong candidate for automation
TC-012	Navigation gate without both sources	Critical business rule; UI state verifiable
TC-013	Data period mismatch modal	Conditional display verifiable with `toBeVisible()`

TC-014	Revenue per Employee KPI	Cross-dataset calculation — high regression risk
TC-015	Labor Spend as % of Revenue KPI	Same + edge case of sub-1% display with decimals
TC-016	EBITDA Margin KPI	Numeric calculation verifiable
TC-017	Comparison banner	Data from both sources present in a single component

5.3 Cases that will NOT be automated

TC	Case	Reason
TC-002	Clipboard paste	Clipboard access in Playwright is fragile and OS-dependent — better covered manually
TC-009	Automatic SEC EDGAR fetch	External API dependency; prone to rate limits and schema changes. Mocked in backend unit tests
TC-010	Verify most recent 10-K	Relies on real-world data that changes over time — not stable as an automated assertion
TC-018	Seniority pyramid (visual)	Chart.js canvas is not verifiable via standard DOM assertions
TC-019	Value Creation Opportunities	Requires specific dataset conditions; better covered manually with controlled data
TC-020	PowerPoint export	Binary file — limited to verifying the download occurs, not visual content

5.4 Suggested Playwright project structure

```
tests/
├── fixtures/
│   ├── WorkshopRoster.xlsx
│   └── financials-manual.json
├── e2e/
│   ├── roster-analysis.spec.ts
│   ├── financial-data.spec.ts
│   └── integrated-analysis.spec.ts
├── helpers/
│   └── setup.ts
└── playwright.config.ts
```

5.5 Browsers in the Playwright suite

Browser	Priority
Google Chrome	**High**
Microsoft Edge	**High**
Mozilla Firefox	Medium

6. API Testing (Backend)

6.1 POST /api/roster/analyze

Case	Input	Expected result
Happy path	Valid Excel with all required columns	200 + JSON with all aggregation fields
Missing column	Excel without `Seniority` column	422 with message indicating the missing column
Invalid format	`.pdf` file	400 with format error message

Alternative column names	Excel with `"Seniority Level"` instead of `"Seniority"`	200 — fuzzy matching should resolve it

6.2 GET /api/financials/{ticker}

Case	Input	Expected result
Valid ticker	`AAPL`	200 + JSON with most recent 10-K data
Invalid ticker	`XXXXX`	404 with clear message
SEC EDGAR unavailable	(mocked timeout)	502 with fallback message to manual entry

7. Entry and Exit Criteria

7.1 Entry Criteria

- Local environment is running and accessible
- All 20 happy path TCs have a corresponding implementation
- `WorkshopRoster.xlsx` is available in `QA/`
- Playwright tests are configured and run without setup errors

7.2 Exit Criteria

- 100% of happy path TCs (TC-001 to TC-020) passed
- Playwright automated tests running in CI without flakiness
- All API endpoints respond correctly for happy path cases
- No visual regressions on main screens compared to last approved version

8. Risks and Mitigations

Risk	Likelihood	Impact	Mitigation

SEC EDGAR changes its response schema	Medium	High	Mock the API in E2E tests; backend must have robust error handling
Chart.js graphs render incorrectly in some browsers	Medium	Medium	Manual verification on each target browser
Fuzzy column matching fails with unexpected column name variants	High	High	Document and test all known column name variants
PPT export generates a corrupted file	Low	High	Verify download + open the file manually on every regression cycle
AI narrative returns inconsistent responses	High	Low	Do not automate — verify the UI handles loading/error states correctly

9. Test Data

File	Description	Location
`WorkshopRoster.xlsx`	Employee roster, 2 countries, 9 functions, FLC column	`QA/WorkshopRoster.xlsx`
Manual financial data (TBD)	JSON with Revenue / EBITDA / Expenses for a test company	`QA/fixtures/` (to be created)
Test tickers	`AAPL` (Apple), `MSFT` (Microsoft)	—

10. Related Documents

Document	Location

Happy Path Test Cases (MD)	`QA/test-cases-happy-path.md`
Happy Path Test Cases (Excel)	`QA/test-cases-happy-path.xlsx`
Test Roster	`QA/WorkshopRoster.xlsx`